

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 411 — Past Questions & Materials
by Renance

Course: MTH 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — MTH 411 THEORY OF ORDINARY DIFFERENTIAL EQUATION 2024/2025

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2024/2025 SESSION
COURSE TITLE & CODE: THEORY OF ORDINARY DIFFERENTIAL EQUATION (MTH 411)
Instruction: Attempt three questions only Time: 2hrs, 30mins

Question 1

- State and prove the Gronwall-Keid-Bellman inequality theorem.
- Given a system of first-order ODE:
$$\begin{cases} y_1' = -y_1 - y_2 - y_1^3 \\ y_2' = -y_1 - y_2 - y_2^3 \end{cases}$$
 Determine the stability of the trivial solution of the ODE.

Question 2

- State and prove the Peano's existence theorem
- Given an ODE $X' = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix} X$, $X(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, where $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$. Solve for X.

Question 3

- Given an IVP, $x' = f(t, x)$, $x(t_0) = x_0$ state the existence and uniqueness theorem.
- (i). Define an integral equation, (ii) when is an integral equation said to be regarded as Fredholm equation of the second kind?
- Show that $f(t, x) = x^2 - t^2$ satisfies the Lipschitz condition on $R = \{(t, x) : 0 \leq t \leq 1, |x - 1| \leq 1\}$ and find the approximating Lipschitz constant.

Question 4

- Give a space $C([a, b], R^n)$ with metric defined by $d(f, g) = \sup \|f(x) - g(x)\|$, show that the space is a complete metric space.
- List and discuss special type of Kernels

Question 5

Solve the Fredholm integral equation of the second kind, $f(x) = x + \lambda \int_0^1 (xt^2 + x^2t) f(t) dt$ to determine the solution.

Paper 2 of 2 — MTH 411 Ordinary Differential Equations 2018 2019 2020 2021

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
B.Sc (MATHEMATICS) DEGREE EXAMINATION (2018/2019), FIRST SEMESTER.
MTH 411 (ORDINARY DIFFERENTIAL EQUATIONS 3C). April, 2019. Duration: 2hrs 30mins.

Instructions: Answer ANY FOUR (4) questions
 All questions carry equal Marks

Question 1

- (a) What is a Lipschitz Function?
 (b) Reduce the following boundary value problem into an integral equation

$$\frac{d^2 y}{dx^2} + \lambda y = 0, \quad y(0) = 0, y(\lambda) = 0.$$

 (c) Let $p(x) > 0$ and $p'(x), r_1(x), r_2(x)$ continuous functions on $I = [a, b]$ and $r_1(x) > r_2(x)$ on I . Further, if $\phi(x)$ is a nontrivial solution of

$$(p(x)y')' + r_1(x)y = 0$$

 and $\psi(x)$ is nontrivial solution of

$$(p(x)y')' + r_2(x)y = 0$$

 then between any two consecutive zeros of $\phi(x)$ there exists a zero of $\psi(x)$

Question 2

- (a) Define the n th iterated kernel
 (b) Find the first 3 terms of the Neumann series for the integral equation

$$u(x) = 1 + \epsilon \int_0^1 (x - y)u(y) dy.$$

where $\epsilon \in R$.

- (c) Suppose that f and $\partial f / \partial x$ are continuous and bounded in a given region U . Let ϕ_{t_0, x_0} and φ_{t_0, x_0} be solutions of

$$x' = f(t, x), x(t_0) = x_0$$

passing through (t_0, x_0) and $(\hat{t}_0, \hat{x}_0)$ respectively. Suppose that ϕ and φ exist on some interval I . Prove that, for each $\epsilon > 0$, there exists $\delta > 0$ such that if $|t - \hat{t}| < \delta$ and $\|x_0 - \hat{x}_0\| < \delta$, then

$$\|\phi(t) - \varphi(t)\| < \epsilon, \text{ for } t, \hat{t} \in I.$$

Question 3

- (a) Define a Monodromy operator associated to

$$x' = A(t)x, A(t + \omega) = A(t), \forall t$$

 with entries of $A(t)$ continuous on R .
 (b) Let $x' = f(t, x)$, when is a function $\phi(t)$ said to be a solution to

$$x' = f(t, x), x(t_0) = x_0$$

 (c) Let $R(t, t_0)$ be the resolvent of the homogeneous equation $x' = A(t)x$ associated to $x' = A(t)x + b(t)$ where $x \in R^n$, $A \in M_n(R)$ and $b \in R^n$, with A and b continuous and ω -periodic. Then the solution $x(t)$ is given by

$$x(t) = R(t, t_0) + \int_{t_0}^t R(t, s)B(s)ds$$